Dimension Theorem

1 Dimension Theorem

Lemma 1 (Replacement Lemma)**.** *Let V be a vector space spanned by n vectors. Then every linearly independent subset of V contains at most n vectors.*

Theorem 1 (Dimension Theorem)**.** *Let V be a finite-dimensional vector space. Then any two bases of V have the same number of elements.*

Proof. Let

$$B_1 = \{v_1, v_2, \ldots, v_m\}$$

and

$$B_2 = \{w_1, w_2, \ldots, w_n\}$$

be two bases of V.

Since B_1 is linearly independent and B_2 spans V, the Replacement Lemma implies that

$$m \leq n.$$

Similarly, since B_2 is linearly independent and B_1 spans V, we obtain

$$n \leq m.$$

Combining these inequalities,

$$m \leq n \quad \text{and} \quad n \leq m,$$

we conclude that

$$\boxed{m = n}.$$

Therefore, any two bases of V have the same number of elements. □

Definition 1 (Dimension)**.** *The dimension of a finite-dimensional vector space V is the number of vectors in any basis of V.*

It is denoted by

$$\dim(V) = n.$$